

# Mirage

X17 (2017) — English translation

On a hot summer day, when the air warmed up to  $t_1 = 35^\circ\text{C}$ , the temperature of the asphalt in the sun and the thin layer of air above it was  $t_2 = 65^\circ\text{C}$ . When walking along the highway in such dry weather, “puddles” are often visible on the road. The refractive index of air depends linearly on air density:  $n = 1 + k\rho$ . At  $t_0 = 0^\circ\text{C}$ ,  $n_0 = 1.000292$ . Consider that a person’s height is  $H = 1.8\text{ m}$ .

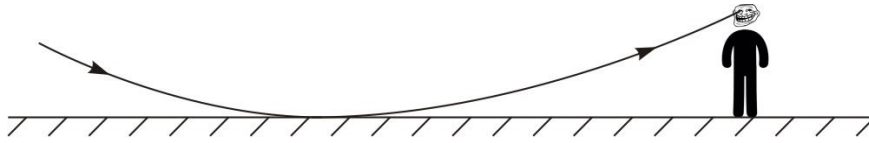


Figure 1: Figure 1.

**A1** At what minimum distance  $L_{min}$  can a “puddle” be seen? Express the answer **5.00** in terms of  $H$ ,  $n_0$ ,  $T_0$ ,  $T_1$ ,  $T_2$  and calculate the resulting value.

Source: <https://pho.rs/p/3942>